ATS Automation - Implementation Plan

Overview

Phase	Timeline	Focus Areas	Clients
Phase 1	Weeks 1-6	Quick Wins: session pop-ups, auto-notes, wrap-up sync, lead pruning, auto-fax, PDF filler	All 6 clients
Phase 2	Weeks 7-14	Intelligence Layer: AI reply/note suggesters, queue prioritizer, negotiation form-filler, portal scraping	All 6 clients
Phase 3	Weeks 15-20	Tuning & Expansion: Model accuracy tuning, new portal profiles, team config UIs, feedback loops	Legacy, Banyan, TBT

Phase 1: Quick Wins (Weeks 1-6)

Goal

Deploy high-impact, lower-complexity automations that deliver immediate time savings.

Automations

1. CTM-SF Auto-Account Access (Flyland)
2. Auto-Note Generator (Flyland)
3. Wrap-Up Auto-Sync (Legacy)
4. Customer History Auto-Pull (Legacy)
5. Auto-Lead Pruning (TBT)
6. Auto-Lead Pull & Inbound Routing (TBT)
7. Auto-Fax for Appeals (Takahami)
8. Claim-Type-Based Appeal Router (Takahami)
9. CTM-SF Auto-Account Pop-Up (Banyan)
10. Auto-Call Tracking & Form/Note Auto-Fill (Banyan)
11. PDF Auto-Filler (Element)

Weekly Breakdown

Week 1-2: Foundation

- [ ] Setup development environment
- [ ] Install Playwright, PyAutoGUI, NLP libraries
- [ ] Create Chrome Extension skeleton
- [ ] Build DOM profile for Flyland (CTM + SF)
- [ ] Test CTM call event detection

Week 3-4: Core Development

- [ ] Implement CTM-SF Auto-Account Access (Flyland)
- [ ] Implement Auto-Note Generator (Flyland)
- [ ] Implement PDF Auto-Filler (Element)
- [ ] Test all automations in dev environment

Week 5-6: Deployment & Testing

- [ ] Deploy to 2-3 pilot agents (Flyland)
- [ ] Collect feedback, fix bugs
- [ ] Deploy remaining Phase 1 automations
- [ ] Document issues for Phase 2

Success Criteria

- Each automation saves 1-3 mins per interaction
- Zero credential exposure
- 80%+ automation success rate

Phase 2: Intelligence Layer (Weeks 7-14)

Goal

Add AI-powered automations for more complex tasks.

Automations

1. Chat Reply Suggester & Form Auto-Filler (Flyland)
2. Form-Filler, Note Summarizer & AI Queue Prioritizer (Legacy)
3. Text Reply Suggester, Note Templates & Insurance Status Checker (TBT)
4. Note Summarizer & Reply Suggester (Banyan)
5. Negotiation Form-Filler & Reply Suggester (Takahami)
6. Pre-Filled Insurance Form Template Generator (Element)

Weekly Breakdown

Week 7-8: NLP Foundation

- [ ] Download and test DistilBERT model
- [ ] Build summarization pipeline
- [ ] Create reply template system
- [ ] Test offline performance

Week 9-10: AI Integration

- [ ] Integrate NLP with Chrome Extension

- [ ] Build overlay panel for suggestions
- [ ] Implement clipboard monitoring
- [ ] Test with real calls (pilot)

Week 11-12: Portal Intelligence

- [ ] Build Availity portal scraper (Element)
- [ ] Build negotiation form-filler (Takahami)
- [ ] Build insurance status checker (TBT)
- [ ] Test with production data

Week 13-14: Refinement

- [ ] Polish all Phase 2 automations
- [ ] Deploy to all agents
- [ ] Collect feedback
- [ ] Plan Phase 3 refinements

Success Criteria

- 30%+ time reduction per task
- <5% error rate on AI suggestions
- Agent satisfaction >4/5

Phase 3: Tuning & Expansion (Weeks 15-20)

Goal

Fine-tune existing automations based on feedback and expand to new scenarios.

Focus Areas

1. Model accuracy tuning based on real outputs
2. New portal profiles (as clients add systems)
3. Team lead configuration UIs
4. Field-mapping updates without code changes
5. User feedback loops

Weekly Breakdown

Week 15-16: Feedback Integration

- [ ] Review all agent feedback
- [ ] Identify top 10 issues
- [ ] Prioritize fixes by impact
- [ ] Update NLP model training

Week 17-18: Config UIs

- [ ] Build field-mapping UI for non-technical users
- [ ] Build template editor UI
- [ ] Build rule configuration UI
- [ ] Test with team leads

Week 19-20: Optimization

- [ ] Performance optimization (reduce memory/CPU)
- [ ] Browser update compatibility fixes
- [ ] Documentation updates
- [ ] Final deployment to all agents

Success Criteria

- <2% automation failure rate
- Non-technical users can update configs
- 50%+ reduction in manual tasks

Rollout Strategy

Pilot Program (Phase 1)

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Week 3: 2-3 agents (Flyland)
Week 5: 5-8 agents (all clients)
Week 6: Full rollout
```

Staggered Client Deployment

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Week 1-2: Flyland (most mature CTM/SF setup)
Week 3-4: Legacy, TBT (similar systems)
Week 5-6: Banyan, Takahami, Element (unique setups)
```

Testing Strategy

Unit Tests

- Each function has >80% test coverage
- Mock DOM elements for isolated testing

Integration Tests

- Test against CTM/SF staging (if available)

- Test Chrome Extension + Playwright integration

User Acceptance Testing

- 5 agents per client
- 2-week UAT period
- Survey-based feedback

Performance Testing

- Memory usage <500MB
- CPU idle when not on call
- <2s response time for overlays

Resource Requirements

Team Structure

Role	Count	Phase 1	Phase 2	Phase 3
Python Developer	2	Full	Full	1
Chrome Extension Dev	1	Full	Part	Part
QA Engineer	1	Full	Full	Part
Project Manager	1	Part	Part	Part

Equipment

- 3 developer machines (8GB RAM minimum)
- 5 test agent machines (for UAT)
- Central server (optional, for config management)

Timeline Summary

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Phase 1 (Quick Wins)      ▓▓▓▓▓▓▓▓▒▒▒▒▒▒▒▒  Weeks 1-6
Phase 2 (Intelligence)    ▒▒▒▒▒▒▒▒▓▓▓▓▓▓▓▓  Weeks 7-14
Phase 3 (Tuning)          ▒▒▒▒▒▒▒▒▒▒▒▒▓▓▓▓  Weeks 15-20

Total Duration: 20 Weeks
```

Go/No-Go Criteria

Phase 1 Go

- [ ] Development environment ready
- [ ] Chrome Extension loads without errors
- [ ] Playwright can navigate CTM/SF
- [ ] DOM selectors identified for Flyland

Phase 2 Go

- [ ] Phase 1 automations deployed
- [ ]

  > 80% success rate achieved

- [ ] NLP models downloaded and tested
- [ ] Agent feedback collected

Phase 3 Go

- [ ] Phase 2 automations deployed
- [ ] <5% error rate
- [ ] Config UIs functional
- [ ] Documentation complete

Risk-Adjusted Timeline

Risk	Likelihood	Impact	Mitigation	Buffer
Portal UI changes	High	Medium	Modular scraper profiles	+1 week
NLP model accuracy	Medium	Medium	Confidence thresholds	+1 week
Agent adoption	Medium	High	Training sessions	+1 week
Browser updates	Medium	Medium	Version pinning	+1 week

Adjusted Timeline: 24 weeks (20 planned + 4 buffer)

Last updated: March 2026